ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** receptor -> sensory neuron -> spinal cord -> motor neuron -> effector	Standard reflex pathway.
2	**(d)** auxin affects stem and root cells in opposite ways	Differential response.
3	**(c)** insulin	Diabetes.
4	**(b)** relay neuron	Relay neuron.
5	**(b)** does not occur	Auxin source is lost.
6	**(b)** growth hormone	Pituitary.
7	**(b)** muscle	Muscle contracts.
8	**(d)** abscisic acid	ABA.
9	**(b)** iodine	Goitre.
10	**(a)** dendrite to axon end	Direction.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Quick response avoids danger (1); Does not wait for brain processing (1).	1+1
12	Chemical is released only at axon terminal (1); Dendrite receives (1).	1+1
13	Regulate elongation, division (1); Coordinate responses (1).	1+1
14	Reflex: involuntary, spinal cord, fast (1); Voluntary: brain controls, slower (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thinking and memory (1); Voluntary actions and speech (1); Senses like sight interpretation (1).	1+1+1
16	Low thyroxine causes pituitary to release more TSH (1); Thyroid makes thyroxine (1); Level returns to normal (1).	1+1+1
17	Hormones travel in blood (1); Need time to reach target (1); Chemical changes take time (1).	1+1+1
	OR: Impulse reaches the axon end (1); Chemical released into the synapse (1); Starts an impulse in the next dendrite (1).	1+1+1
18	Blood diverted to muscles (1); Heart rate increases (1); Skin blood vessels contract (1).	1+1+1
	OR: Climbing plants get support (1); Reach sunlight (1); Differential growth (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Receptor in skin (1); Sensory neuron to spinal cord (1); Relay and motor neuron (1); Effector muscle moves the foot (1); Brain informed later (1).	1+1+1+1+1
	OR: Sugar rises after meal (1); Insulin released (1); Sugar stored/used (1); Diabetes: not enough insulin (1); Insulin injections treat (1).	1+1+1+1+1
20	Cerebrum: thinking, reading (1); Cerebellum: riding a cycle (1); Medulla: salivation (1); Pons: breathing control relay (1); Midbrain: eye movements (1).	1+1+1+1+1
	OR: Dendrite, cell body, axon (1); Impulse is electrical in axon (1); At the axon end, chemical released (1); Chemical diffuses across the synapse (1); Starts an impulse in the next neuron (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.

- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.
- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** cytokinin	Cell division.
2	**(c)** adrenaline	Adrenal gland.
3	**(a)** chemicals	Neurotransmitters.
4	**(c)** nastic movement	Not directional.
5	**(d)** thyroxine	Thyroxine.
6	**(d)** cerebellum	Balance and posture.
7	**(a)** loss of water from cells	Turgor change.
8	**(c)** increase in breathing rate	Fight or flight.
9	**(b)** forebrain	Hunger centre.
10	**(a)** directional	Direction depends on stimulus.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Auxin accumulates on the shaded side (1); More cell elongation there (1).	1+1
12	Lowers blood sugar (1); Replaces deficient insulin (1).	1+1
13	Connect sensory and motor neurons (1); Pass information to the brain too (1).	1+1
14	Tropic: direction depends on stimulus (1); Nastic: independent of direction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nerve: electrical impulses fast (1); Hormone: chemical through blood slow (1); Nerve affects specific organ; hormone far (1).	1+1+1
16	Gibberellin: stem elongation in dwarf plants (1); Auxin: phototropism and cell elongation (1); Both promote growth (1).	1+1+1
17	Shoot bends up and towards light (1); Root bends down (1); Auxin plays the key role (1).	1+1+1
	OR: Tip removed: no bending (1); Tip covered: no bending (1); Auxin from tip controls bending (1).	1+1+1
18	Inhibits growth (1); Causes dormancy (1); Wilting of leaves under stress (1).	1+1+1
	OR: Response by turgor, not growth (1); Water moves rapidly (1); No new cell growth (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Phototropism: auxin on shaded side (1); Shoot bends toward light (1); Geotropism: auxin accumulates on lower side (1); In roots, higher auxin inhibits growth (1); Root bends down (1).	1+1+1+1+1
	OR: Iodine needed for thyroxine (1); Thyroxine controls metabolism (1); Deficiency causes goitre (1); Source: iodised salt (1); Symptoms: swelling in neck (1).	1+1+1+1+1
20	Insulin: diabetes (1); Thyroxine: goitre (1); Growth hormone: dwarfism (1); Adrenaline: stress response (1); Testosterone: puberty in males (1).	1+1+1+1+1
	OR: Adrenal glands secrete adrenaline (1); Heart rate rises (1); Breathing rate rises (1); Blood to the muscles (1); Nerves send immediate signals (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.

- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** puberty	Testes and ovaries.
2	**(c)** medulla	Hindbrain.
3	**(d)** faster growth on the side away from contact	Differential growth.
4	**(c)** iodine	Iodine.
5	**(d)** spinal cord	Reflex.
6	**(d)** hydrotropism	Hydrotropism.
7	**(a)** pancreas	Exocrine and endocrine.
8	**(b)** from receptors to the spinal cord/brain	Sensory neuron.
9	**(c)** diffusion from cell to cell	No circulatory system.
10	**(b)** They are carried by nerves	Hormones travel in blood.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	By blood sugar level (1); Feedback mechanism (1).	1+1
12	Detects danger via receptors (1); Sends quick signals to effectors (1).	1+1
13	Water moves out of pulvinus cells (1); No, it is a turgor change (1).	1+1
14	Cerebrum: thinking and voluntary actions (1); Cerebellum: balance and precise movement (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Roots grow to water (1); Grow down into soil (1); Anchor plant and absorb water (1).	1+1+1
16	Both have dendrite, cell body, axon (1); Sensory carries signals to CNS (1); Motor carries signals from CNS (1).	1+1+1
17	Receives information (1); Processes and decides (1); Sends instructions to effectors (1).	1+1+1
	OR: Controls other glands (1); Growth hormone (1); TSH and others (1).	1+1+1
18	Signals between body and brain are blocked (1); Loss of movement or feeling (1); Reflex arcs may be lost (1).	1+1+1
	OR: Under: dwarfism (1); Over: gigantism (1); Produced by the pituitary (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fast vs slow (1); Neurons vs blood (1); Short vs long (1); Immediate response vs long-term regulation (1); Together coordinate the body (1).	1+1+1+1+1
	OR: Use chemicals (hormones) (1); Change in turgor of cells (1); Auxin in phototropism (1); Mimosa leaf movement (1); Slower than animals (1).	1+1+1+1+1
20	Tropic: directional (1); Nastic: non-directional (1); Hormones: auxin, cytokinin (1); Touch: Mimosa (1); No nervous system in plants (1).	1+1+1+1+1
	OR: Pathway between body and brain (1); Reflex centre (1); Vertebral column (1); Meninges and CSF (1); Skull for brain (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
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- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

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